Problem H2: The Dirac Current $\bar{u}\gamma^\mu u = Kp^\mu$

Proved Explicitly from Griffiths' Spinor Solutions

Based on Griffiths §7.2 / Thomson §4.6

Particle Physics Study Guide

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1 Problem Statement

Problem Statement

H2) Show that $\bar{u}(p_1)\gamma^\mu u(p_1) = K\, p_1^\mu$ where K is a constant. **Don't just use the result of H3**, but rather show it explicitly using the forms of the $u/\bar{u}$ solutions given in Griffiths. We will probably do this in class together.

What This Problem Wants

Take the *explicit* Griffiths spinors $u^{(1)}$ and $u^{(2)}$ (Eq. 7.46), construct $\bar{u} = u^\dagger\gamma^0$, and **directly compute** $\bar{u}\gamma^\mu u$ component by component ($\mu = 0, 1, 2, 3$). Show that the result is proportional to $p^\mu = (E, p_x, p_y, p_z)$ with the same constant K for every component.

What you are NOT allowed to do: Use the Gordon identity (presumably H3), completeness relations, or any trace trick. The point is to get your hands dirty with the actual 4×4 matrix multiplication.

2 Prerequisites: Everything You Need Before Starting

2.1 What is a Dirac Spinor?

The Big Picture

In non-relativistic quantum mechanics, a spin-$\frac{1}{2}$ particle is described by a 2-component spinor ("spin up" or "spin down"). Relativistic quantum mechanics (the Dirac equation) requires **4 components**: two for spin and two for particle vs. antiparticle degrees of freedom.

A free-particle solution to the Dirac equation is:

$$\psi(x,t) = u(E,\mathbf{p})\, e^{i(\mathbf{p}\cdot\mathbf{x} - Et)}$$

where $u(E,\mathbf{p})$ is a 4-component **Dirac spinor**—a column vector that encodes the spin state and is a function of the particle's energy and momentum.

2.2 Griffiths' Explicit Spinors (Eq. 7.46)

The Spinor Solutions $u^{(1)}$ and $u^{(2)}$

In the Dirac–Pauli representation, the two positive-energy particle spinors are (natural units, $c = \hbar = 1$):

$$u^{(1)} = N \begin{pmatrix} 1 \\ 0 \\ \dfrac{p_z}{E+m} \\ \dfrac{p_x + ip_y}{E+m} \end{pmatrix}, \qquad u^{(2)} = N \begin{pmatrix} 0 \\ 1 \\ \dfrac{p_x - ip_y}{E+m} \\ \dfrac{-p_z}{E+m} \end{pmatrix} \tag{1}$$

where $N = \sqrt{E+m}$ is the normalization factor (giving $u^\dagger u = 2E$), and $E^2 = |\mathbf{p}|^2 + m^2$.

Reading the Spinor: Upper vs. Lower Components

Each spinor naturally splits into a "big" 2-component piece u_A and a "small" piece u_B:

$$u = \begin{pmatrix} u_A \\ u_B \end{pmatrix}, \qquad u_B = \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} u_A$$

For $u^{(1)}$: $u_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (spin up along z).

For $u^{(2)}$: $u_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (spin down along z).

The lower component u_B is related to u_A by the "$\boldsymbol{\sigma} \cdot \mathbf{p}$" operator—this is how **spin and momentum get entangled** in relativistic QM. In the non-relativistic limit ($|\mathbf{p}| \ll m$), u_B is small compared to u_A (suppressed by $\sim |\mathbf{p}|/2m$).

2.3 The $\boldsymbol{\sigma} \cdot \mathbf{p}$ Matrix

$\boldsymbol{\sigma} \cdot \mathbf{p}$ in Components

The Pauli matrices are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Their contraction with $\mathbf{p}$ is:

$$\boldsymbol{\sigma} \cdot \mathbf{p} = \sigma_x p_x + \sigma_y p_y + \sigma_z p_z = \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix} \tag{2}$$

Key identity (from the Pauli algebra $\sigma_i \sigma_j = \delta_{ij} I + i\epsilon_{ijk}\sigma_k$):

$$(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = |\mathbf{p}|^2 \, I \tag{3}$$

2.4 The Adjoint Spinor $\bar{u}$

Definition of $\bar{u}$

The **Dirac adjoint** (or "bar") of a spinor u is defined as:

$$\bar{u} \equiv u^\dagger \gamma^0 \tag{4}$$

where $u^\dagger$ is the conjugate transpose (row vector of complex conjugates), and γ^0 is the diagonal matrix:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

What γ^0 does: it flips the sign of the lower two components. So if $u = \begin{pmatrix} u_A \\ u_B \end{pmatrix}$, then:

$$\bar{u} = u^\dagger \gamma^0 = \begin{pmatrix} u_A^\dagger & -u_B^\dagger \end{pmatrix} \tag{5}$$

Why not just $u^\dagger$? Because $u^\dagger u$ is not Lorentz-invariant—it equals $2E$, which transforms under boosts. The combination $\bar{u}u = u^\dagger \gamma^0 u$ *is* Lorentz-invariant (it equals $2m$). The γ^0 compensates for the fact that boosts don't preserve $u^\dagger u$.

2.5 The Gamma Matrices

The γ^μ in Block Form

In the Dirac–Pauli representation:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^k = \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix} \quad (k = 1, 2, 3) \tag{6}$$

where $\sigma^1 = \sigma_x$, $\sigma^2 = \sigma_y$, $\sigma^3 = \sigma_z$.

3 The Calculation: Compact 2×2 Block Method

Rather than multiplying out all sixteen entries of 4×4 matrices, we use the 2×2 block structure. This is the standard technique and the one your instructor almost certainly intends.

Setup: Write Everything in Blocks

For either spin state $s = 1, 2$, Griffiths' spinor has the form:

$$u^{(s)} = N \begin{pmatrix} \phi_s \\ \dfrac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \phi_s \end{pmatrix} \tag{7}$$

where $\phi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\phi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, and $N = \sqrt{E + m}$.

The adjoint spinor (from Eq. 5) is:

$$\bar{u}^{(s)} = N \begin{pmatrix} \phi_s^\dagger & -\phi_s^\dagger \dfrac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E + m} \end{pmatrix} \tag{8}$$

(using $(\boldsymbol{\sigma} \cdot \mathbf{p})^\dagger = \boldsymbol{\sigma} \cdot \mathbf{p}$ since $\sigma_i^\dagger = \sigma_i$ and p_i are real.)

3.1 Computing $\bar{u}\gamma^0 u$ (the $\mu = 0$ component)

Using $\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$:

$$\bar{u}\gamma^0 u = N^2 \begin{pmatrix} \phi^\dagger & -\phi^\dagger \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} \phi \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi \end{pmatrix} \tag{9}$$

First, γ^0 acts on the column:

$$\gamma^0 \begin{pmatrix} \phi \\ \frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi \end{pmatrix} = \begin{pmatrix} \phi \\ -\frac{\boldsymbol{\sigma} \cdot \mathbf{p}}{E+m} \phi \end{pmatrix}$$

Now dot with the row:

$$\bar{u}\gamma^0 u = N^2 \left[\phi^\dagger \phi + \phi^\dagger \frac{(\boldsymbol{\sigma} \cdot \mathbf{p})^2}{(E+m)^2} \phi \right] \tag{10}$$

Use $(\boldsymbol{\sigma} \cdot \mathbf{p})^2 = |\mathbf{p}|^2 I$ and $\phi^\dagger \phi = 1$:

$$= N^2 \left[1 + \frac{|\mathbf{p}|^2}{(E+m)^2} \right] = (E + m) \cdot \frac{(E+m)^2 + |\mathbf{p}|^2}{(E+m)^2} \tag{11}$$

Now use $E^2 = |\mathbf{p}|^2 + m^2$, so $|\mathbf{p}|^2 = E^2 - m^2$:

$$(E+m)^2 + |\mathbf{p}|^2 = E^2 + 2Em + m^2 + E^2 - m^2 = 2E^2 + 2Em = 2E(E+m) \tag{12}$$

Therefore:

$$\bar{u}\gamma^0 u = (E+m) \cdot \frac{2E(E+m)}{(E+m)^2} = 2E \tag{13}$$

Result: $\mu = 0$ Component

$$\boxed{\bar{u}\gamma^0 u = 2E = 2\,p^0} \tag{14}$$

3.2 Computing $\bar{u}\gamma^k u$ (the $\mu = k$ spatial components)

Using $\gamma^k = \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix}$:

$$\bar{u}\gamma^k u = N^2 \begin{pmatrix} \phi^\dagger & -\phi^\dagger \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m} \end{pmatrix} \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix} \begin{pmatrix} \phi \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\phi \end{pmatrix} \tag{15}$$

First, γ^k acts on the column:

$$\gamma^k \begin{pmatrix} \phi \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\phi \end{pmatrix} = \begin{pmatrix} \frac{\sigma^k(\boldsymbol{\sigma}\cdot\mathbf{p})}{E+m}\phi \\ -\sigma^k\phi \end{pmatrix}$$

Now dot with the row:

$$\bar{u}\gamma^k u = N^2 \left[\phi^\dagger \frac{\sigma^k(\boldsymbol{\sigma}\cdot\mathbf{p})}{E+m}\phi + \phi^\dagger \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})\sigma^k}{E+m}\phi \right] \tag{16}$$

Factor out $N^2/(E+m)$:

$$\bar{u}\gamma^k u = \frac{N^2}{E+m}\, \phi^\dagger \Big[\sigma^k(\boldsymbol{\sigma}\cdot\mathbf{p}) + (\boldsymbol{\sigma}\cdot\mathbf{p})\sigma^k \Big] \phi \tag{17}$$

The Key Step: Pauli Anticommutator

The expression in brackets is the **anticommutator** $\{\sigma^k, \boldsymbol{\sigma}\cdot\mathbf{p}\}$. Using the Pauli algebra:

$$\sigma^i\sigma^j = \delta^{ij} I + i\epsilon^{ijk}\sigma^k$$

the anticommutator is:

$$\{\sigma^k, \sigma^j\} = \sigma^k\sigma^j + \sigma^j\sigma^k = 2\delta^{kj} I \tag{18}$$

Contracting with p_j (i.e., replacing $\sigma^j \to \boldsymbol{\sigma}\cdot\mathbf{p} = \sigma^j p_j$):

$$\sigma^k(\boldsymbol{\sigma}\cdot\mathbf{p}) + (\boldsymbol{\sigma}\cdot\mathbf{p})\sigma^k = 2p^k\, I \tag{19}$$

This is the heart of the proof. The cross terms (the $i\epsilon^{ijk}\sigma^k$ pieces) cancel in the anticommutator because the Levi-Civita symbol is antisymmetric, leaving only the δ^{kj} part.

Substituting Eq. (19) into Eq. (17):

$$\bar{u}\gamma^k u = \frac{N^2}{E+m}\, \phi^\dagger\, (2p^k\, I)\, \phi = \frac{N^2}{E+m} \cdot 2p^k \cdot \underbrace{\phi^\dagger\phi}_{=1} \tag{20}$$

Since $N^2 = E + m$:

$$\bar{u}\gamma^k u = \frac{(E+m)}{(E+m)} \cdot 2p^k = 2p^k \tag{21}$$

Result: Spatial Components

$$\boxed{\bar{u}\gamma^k u = 2p^k \qquad (k = 1, 2, 3)} \tag{22}$$

4 Combining the Results

Collecting the $\mu = 0$ result (Eq. 14) and the $\mu = k$ results (Eq. 22):

Final Result

$$\boxed{\bar{u}(p)\,\gamma^\mu\,u(p) = 2p^\mu} \tag{23}$$

where $p^\mu = (E, p_x, p_y, p_z)$. The constant is $K = 2$.

This holds for **both** spin states $u^{(1)}$ and $u^{(2)}$ individually, because the derivation only used $\phi^\dagger\phi = 1$, which is true for both $\phi_1 = \begin{pmatrix}1\\0\end{pmatrix}$ and $\phi_2 = \begin{pmatrix}0\\1\end{pmatrix}$.

Why This Result Makes Physical Sense

The quantity $j^\mu = \bar{\psi}\gamma^\mu\psi$ is the **Dirac probability current** (Griffiths §7.4 / Thomson §4.5.1). For a free particle:

$$j^\mu = \bar{u}\gamma^\mu u = 2p^\mu$$

- The $\mu = 0$ component is $j^0 = 2E$, which is the **probability density** (normalized to $2E$ particles per unit volume in the standard convention).
- The $\mu = k$ components are $j^k = 2p^k$, which is the **probability current**—particles carry their momentum along with them.

Since the current must be a 4-vector (the Dirac equation is Lorentz-covariant), and the *only* 4-vector available for a free particle with definite momentum is p^μ itself, we *must* have $\bar{u}\gamma^\mu u = Kp^\mu$ for some constant. The explicit calculation shows $K = 2$.

5 The Proof at a Glance (Summary)

Step	What You Do	Key Identity Used
1	Write $u = N\begin{pmatrix}\phi \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\phi\end{pmatrix}$, $\bar{u} = N\begin{pmatrix}\phi^\dagger & -\phi^\dagger\frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\end{pmatrix}$	Definition of $\bar{u} = u^\dagger\gamma^0$
2	Compute $\bar{u}\gamma^0 u$	$(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = \lvert\mathbf{p}\rvert^2$, $E^2 - m^2 = \lvert\mathbf{p}\rvert^2$
3	Get $\bar{u}\gamma^0 u = 2E = 2p^0$	Algebra
4	Compute $\bar{u}\gamma^k u$	$\gamma^k = \begin{pmatrix}0 & \sigma^k \\ -\sigma^k & 0\end{pmatrix}$
5	Recognize anticommutator $\{\sigma^k, \boldsymbol{\sigma}\cdot\mathbf{p}\}$	$\{\sigma^i, \sigma^j\} = 2\delta^{ij}$
6	Get $\bar{u}\gamma^k u = 2p^k$	$\phi^\dagger\phi = 1$
7	Combine: $\bar{u}\gamma^\mu u = 2p^\mu$	QED $\square$

6 Common Mistakes and Pitfalls

1. **Forgetting the sign from γ^0 in $\bar{u}$.** The adjoint spinor is $\bar{u} = u^\dagger\gamma^0$, *not* $u^\dagger$. The γ^0 flips the sign on the lower components. If you forget this, you'll get the wrong relative sign between the two terms in the $\mu = k$ calculation.

2. **Confusing $u^\dagger u$ with $\bar{u}u$.** They are different:

$$u^\dagger u = 2E \quad \text{(not Lorentz invariant)}, \qquad \bar{u}u = 2m \quad \text{(Lorentz invariant)}$$

   Note: $\bar{u}u = \bar{u}\gamma^0 u \cdot \frac{m}{E}$... actually, you can check $\bar{u}u$ directly by computing $\bar{u}\cdot u$ as a row-times-column product (try it!).

3. **Not recognizing the anticommutator.** In step 5, you get $\sigma^k(\boldsymbol{\sigma}\cdot\mathbf{p}) + (\boldsymbol{\sigma}\cdot\mathbf{p})\sigma^k$. Some students try to expand this into component-by-component products. Much cleaner: recognize it as $\{\sigma^k, \sigma^j\}p_j = 2\delta^{kj}p_j = 2p^k$.

4. **Thinking the result depends on the spin state.** It doesn't. The calculation only ever uses $\phi^\dagger\phi = 1$, never the specific form of ϕ. Both spin-up and spin-down give the same current $2p^\mu$.

7 Bonus: Verification for $\bar{u}u = 2m$

As a consistency check, let's verify that $\bar{u}u$ (no γ^μ in between) gives the Lorentz-invariant result $2m$.

$$\begin{aligned}
\bar{u}u &= N^2 \begin{pmatrix} \phi^\dagger & -\phi^\dagger \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m} \end{pmatrix} \begin{pmatrix} \phi \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\phi \end{pmatrix} \\
&= N^2 \left[\phi^\dagger\phi - \phi^\dagger \frac{(\boldsymbol{\sigma}\cdot\mathbf{p})^2}{(E+m)^2}\phi \right] = (E+m)\left[1 - \frac{|\mathbf{p}|^2}{(E+m)^2}\right] \\
&= \frac{(E+m)^2 - |\mathbf{p}|^2}{E+m} = \frac{E^2 + 2Em + m^2 - (E^2 - m^2)}{E+m} = \frac{2m^2 + 2Em}{E+m} \qquad (24)
\end{aligned}$$

$$= \frac{2m(m+E)}{E+m} = 2m \qquad \checkmark$$

Contrast: $u^\dagger u = 2E$ vs. $\bar{u}u = 2m$

The **sign difference** between $\bar{u}\gamma^0 u$ and $\bar{u}u$ arises because γ^0 in $\bar{u}\gamma^0 u$ introduces an *extra* sign flip on the lower components, turning the minus sign in $\bar{u}u$ into a plus sign. This is why $u^\dagger u = \bar{u}\gamma^0 u = 2E$ (plus sign $\Rightarrow$ adds $|\mathbf{p}|^2$ term) while $\bar{u}u = 2m$ (minus sign $\Rightarrow$ subtracts $|\mathbf{p}|^2$ term).

8 Practice Questions

1. Verify $\bar{u}\gamma^\mu u = 2p^\mu$ by brute force for $u^{(1)}$ with $\mathbf{p} = (0, 0, p_z)$ (momentum along z). Write out all four components of $u^{(1)}$ explicitly, compute $\bar{u}^{(1)}$, and multiply out the 4×4 matrices for each γ^μ.

2. Show that $\bar{u}^{(1)}\gamma^\mu u^{(2)} = 0$ for a particle moving along the z-axis. What does this mean physically? [Hint: the current between different spin states vanishes for this momentum direction.]

3. Compute $\bar{v}\gamma^\mu v$ for the antiparticle spinors $v^{(1)}$ (Griffiths Eq. 7.47). Do you also get $2p^\mu$? [Hint: yes, but be careful with signs—recall $\bar{v}v = -2m$.]

4. Starting from $(\gamma^\mu p_\mu - m)u = 0$, *derive* $\bar{u}\gamma^\mu u = 2p^\mu$ without using explicit spinors. [This is presumably what H3 is about—the Gordon identity approach.]

5. Show that $\bar{u}\gamma^5 u = 0$ for a particle with mass $m \neq 0$ at rest ($\mathbf{p} = 0$). What happens in the massless limit?